

Adaptive Attention Pruning for Real-Time Neural Inference on Edge Hardware*

Elena Rostova¹[0000-0002-1825-0097], Marcus Vance²[0000-0001-9234-5678],
and Sofia Chen³[0000-0003-4567-8901]

¹ Nexa AI Research, San Francisco CA 94107, USA

e.rostova@nexa.ai

² Synapse Institute of Technology, Cambridge MA 02139, USA

m.vance@synapse-inst.org

³ Vertex Autonomous Systems Lab, Zurich, Switzerland

s.chen@vertex-lab.ch

Abstract. Deploying Transformer-based architectures on resource-constrained edge computing hardware presents severe challenges due to the quadratic computational complexity of self-attention mechanisms. Existing compression methods, such as static pruning and post-training quantization, often fail to dynamically adapt to varying input complexity, leading to sub-optimal latency-accuracy trade-offs. In this paper, we introduce **AP-Edge**, an adaptive attention pruning framework tailored for real-time inference on edge processors. AP-Edge computes input-dependent dynamic sparsity masks by evaluating query-key magnitude distributions prior to full matrix multiplication. Our experimental evaluation on embedded accelerators demonstrates that AP-Edge achieves up to a $2.4\times$ speedup over unpruned baselines while preserving 98.6% of baseline accuracy across standard benchmarks.

Keywords: Edge Computing · Neural Pruning · Efficient Transformers · Real-Time Inference · Adaptive Sparsity.

1 Introduction

Transformer architectures have established state-of-the-art performance across computer vision, natural language processing, and spatial perception domains. However, their deployment on micro-controllers and embedded systems remains bottlenecked by strict memory bandwidth and energy consumption constraints [3]. The primary compute burden stems from the multi-head self-attention (MHSA) module, where sequence length N yields an $\mathcal{O}(N^2)$ time and spatial overhead.

To mitigate these resource demands, model compression techniques such as magnitude-based weight pruning and post-training quantization have been widely studied. While effective for feed-forward layers, static pruning cannot

* Supported by Nexa AI Research Foundation Grant #2025-ET-892.